

# VPS BASEKIT (VBK) GENERATION 2

## DATA SHEET



Figure 1: Gen2 VBK (MAC, UICIP & PMDS)

The **Vehicle Protection Systems (VPS) Base Kit (VBK)** is designed to provide a Modular Open Systems Architecture (MOSA) within the Survivability domain for U.S. Army ground vehicle platforms. It includes the **Modular Active Controller (MAC)**, the **User Interface Control Panel (UICP)** and the **Power Management Distribution System (PMDS)**, enabling modularity, scalability, and reconfigurability for layered capabilities. The VBK supports rapid development for specific missions, platforms, and threats, reducing costs and increasing soldier survivability.

### MODULAR ACTIVE CONTROLLER (MAC)

Provides control and coordination for the Vehicle Protection System (VPS).

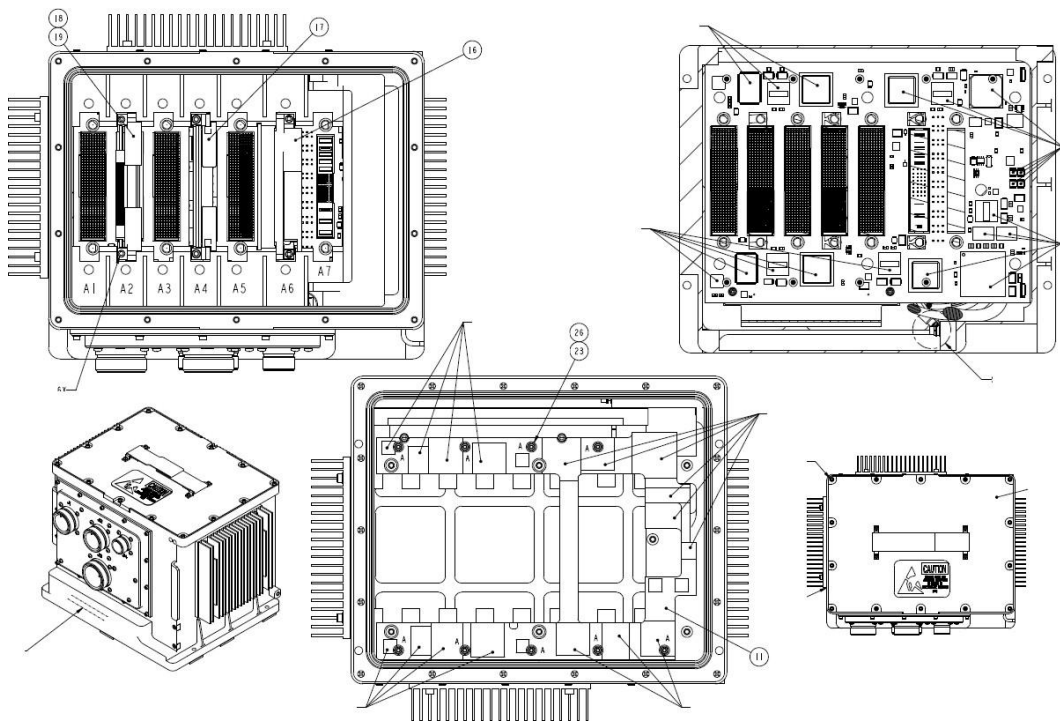


Figure 2: MAC Isometric Views

### FEATURES:

- Open, modular, scalable, and reconfigurable design
- Safety-certifiable hardware and software
- Interfaces with sensors, countermeasures, and other subsystems
- Primary processor – Intel 12 Core Processor (XES 7672) 3U VPX

Supports deterministic Ethernet protocols (AVB)

## SPECIFICATIONS:

**Data Throughput:** 10 Gbps

**Weight:** 35 lbs

**Power Draw:**

Steady State: 250W @ 28VDC (Objective)

Peak: 500W @ 28VDC

Power quality: MIL-STD 1275E

**Dimensions:**

Height: 8.52 inches

Width: 11.83 inches

Depth: 9.79 inches

**Interfaces:**

(16) AVB Capable 1-Gigabit Ethernet Interfaces

(1) Serial Debug Interface

(4) RS485 Interfaces

(4) Bidirectional RS422 Interfaces

(8) Low Speed Opto-Isolated I/O

(12) High Speed Opto-Isolated I/O

(2) CAN 2.0b Bus ISO Interface

(1) GPS Coaxial I/F

## USER INTERFACE CONTROL PANEL (UICP)

Provides vehicle crew members with control interface and status capabilities for the VPS.

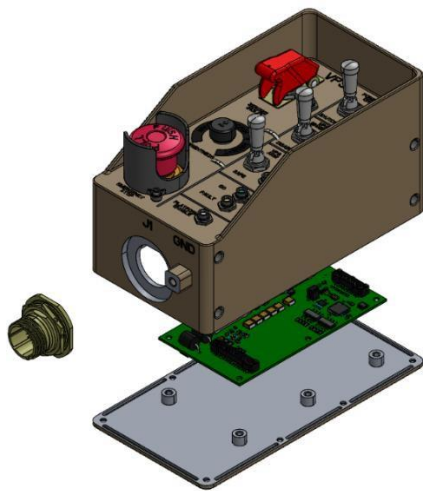


Figure 3: UICP Exploded View

## SPECIFICATIONS:

**Weight:** 4.64 lbs

**Power Draw:**

Steady State: 50W @ 28VDC

Peak: 100W @ 28VDC

**Dimensions:**

Height: 7.50 inches

Width: 4.000 inches

Depth: 4.125 inches

**Interfaces:**

MIL-DTL-38999 connectors for electrical signals

Discrete signals for system control and feedback

CAN Bus (2 channels), RS232 and Ethernet for data exchange

Mounting provisions for integration into vehicle platform

## POWER MANAGEMENT DISTRIBUTION SYSTEM (PMDS)

Provides power distribution and control for VPS sensor and effector subsystems via CAN interface to the MAC



Figure 4: PMDS LRU

### FEATURES:

- 17 Power Channels
- Remote configuration capable
- MIL-STD-1275E, MIL-STD-810F & AOP-53 Safety Certified

### SPECIFICATIONS:

**Weight:** 22 lbs

**Power Draw:**

Steady State: 50W @ 28VDC with no power draw

Peak: 100W @ 28VDC with no power draw

**Dimensions:**

Height: 3.245 inches

Width: 5.524 inches

Depth: 10.2 inches

### Interfaces:

(12) Low Power Channels, 7.5A max

(5) Medium Power Channels, 15A max

(2) High Power Channels, 30A max

7 Discrete inputs

2 Discrete outputs

CAN 2.0B Bus Interface

Mounting provisions for integration into vehicle platform

## SYSTEM SPECIFICATION

### OPERATIONAL STATES & MODES

**Off Mode:** Default state when vehicle power is not supplied; all output discretes are inactive

**On Mode:** System is operational; includes sub-modes such as initialization, standby, active enabled, and fault.

### SAFETY FEATURES

**Emergency Stop:** Terminates power to all subsystems

**Zeroization:** Clears all volatile and non-volatile data storage

**Interlocks:** Prevent accidental initiation of hazardous actions

### ENVIRONMENTAL SPECIFICATIONS

**Operating Temperature:** -40°F to 160°F

**Storage Temperature:** -60°F to 160°F

**Altitude:**

Operating: 1300 feet below and 15,000 feet above sea level

Non-Operating: Up to 50,000 feet above sea level

**Shock and Vibration:**

Ballistic Shock: Continuous operation during exposure to shock impulses

Functional Shock: Exposure to 15 ± 3 g, 75.0ms half sine wave

**Electromagnetic Compatibility:**

MIL-STD-461F and MIL-STD-464C compliance

## SYSTEM INTERFACES

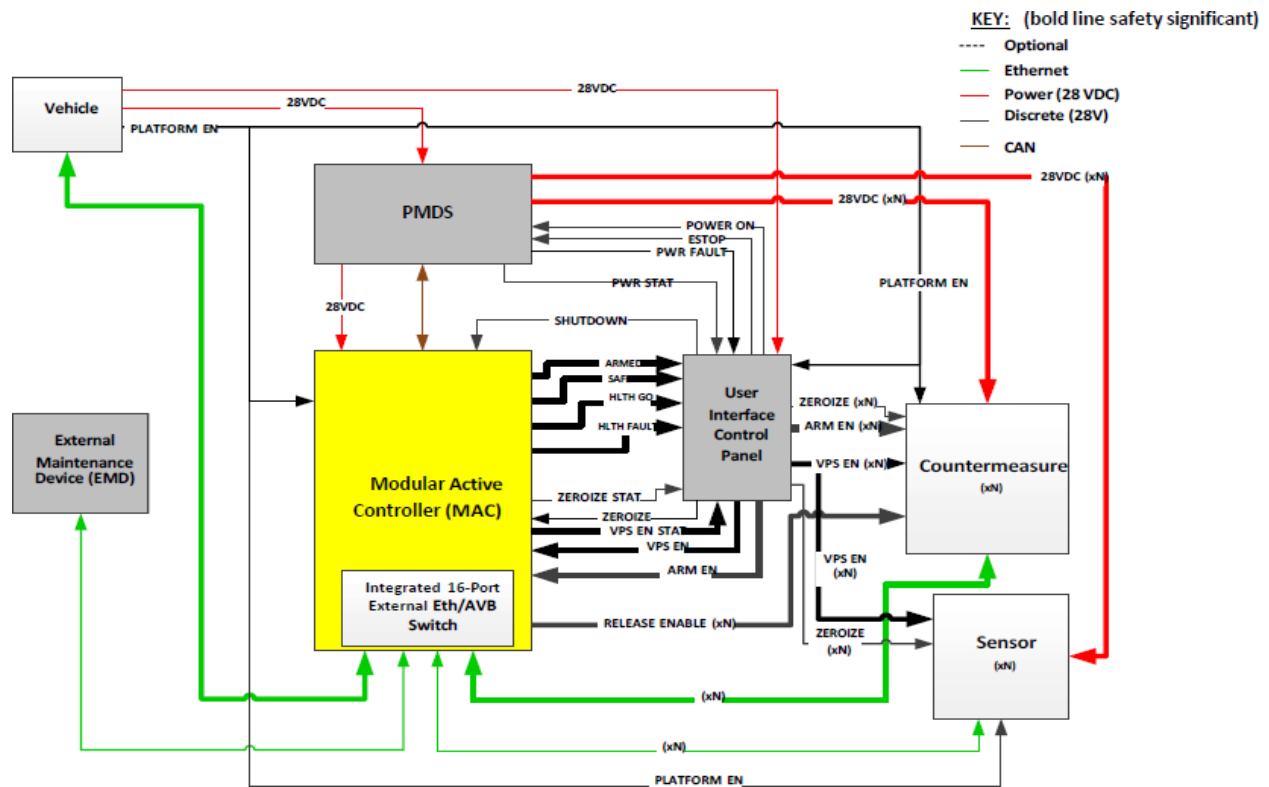


Figure 5: System Interconnect Diagram

## RELIABILITY, AVAILABILITY, MAINTAINABILITY

**Mean Time Between Failures (MTBF):**  $\geq 3,500$  hours

**Mean Time to Repair (MTTR):**  $\leq 0.5$  hours

### Maintenance Ratio:

Field Level:  $\leq 0.005$  maintenance man-hours per operating hour

Sustainment Level:  $\leq 0.001$  maintenance man-hours per operating hour